02/06/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

PROJECT PHOENIX FOR RE-NEET-2026

MM: 720 **CST - 04** **Time: 180 Mins.**

Answers

1. (1)	37. (2)	73. (3)	109. (2)	145. (1)
2. (1)	38. (1)	74. (4)	110. (2)	146. (1)
3. (2)	39. (1)	75. (2)	111. (3)	147. (2)
4. (4)	40. (1)	76. (2)	112. (1)	148. (1)
5. (1)	41. (2)	77. (4)	113. (2)	149. (2)
6. (2)	42. (2)	78. (1)	114. (2)	150. (3)
7. (1)	43. (4)	79. (3)	115. (3)	151. (4)
8. (4)	44. (2)	80. (1)	116. (3)	152. (2)
9. (1)	45. (4)	81. (2)	117. (2)	153. (2)
10. (1)	46. (4)	82. (2)	118. (1)	154. (4)
11. (4)	47. (1)	83. (2)	119. (1)	155. (3)
12. (3)	48. (2)	84. (2)	120. (2)	156. (3)
13. (2)	49. (2)	85. (2)	121. (2)	157. (3)
14. (3)	50. (4)	86. (1)	122. (4)	158. (2)
15. (1)	51. (2)	87. (1)	123. (4)	159. (4)
16. (4)	52. (1)	88. (4)	124. (2)	160. (4)
17. (1)	53. (4)	89. (3)	125. (3)	161. (2)
18. (4)	54. (1)	90. (2)	126. (4)	162. (2)
19. (4)	55. (3)	91. (3)	127. (1)	163. (4)
20. (3)	56. (2)	92. (4)	128. (2)	164. (3)
21. (4)	57. (4)	93. (3)	129. (3)	165. (4)
22. (2)	58. (2)	94. (4)	130. (2)	166. (3)
23. (3)	59. (1)	95. (3)	131. (4)	167. (1)
24. (2)	60. (4)	96. (4)	132. (4)	168. (1)
25. (1)	61. (2)	97. (1)	133. (3)	169. (4)
26. (3)	62. (2)	98. (3)	134. (3)	170. (4)
27. (1)	63. (4)	99. (2)	135. (2)	171. (3)
28. (3)	64. (3)	100. (2)	136. (4)	172. (1)
29. (4)	65. (1)	101. (3)	137. (3)	173. (3)
30. (4)	66. (2)	102. (3)	138. (3)	174. (2)
31. (3)	67. (1)	103. (3)	139. (3)	175. (4)
32. (1)	68. (1)	104. (1)	140. (1)	176. (3)
33. (4)	69. (2)	105. (3)	141. (4)	177. (1)
34. (4)	70. (1)	106. (1)	142. (2)	178. (4)
35. (2)	71. (1)	107. (1)	143. (2)	179. (3)
36. (1)	72. (4)	108. (1)	144. (1)	180. (2)

Hints and Solutions

PHYSICS

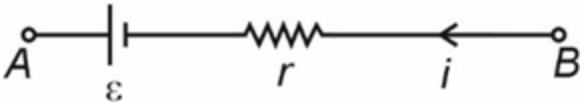
(1) **Answer :** (1)
Solution:
$\varepsilon = \overrightarrow{l} \cdot \left(\overrightarrow{v} \times \overrightarrow{B} \right)$

(2) **Answer :** (1)
Solution:
Both statements A and B are correct.

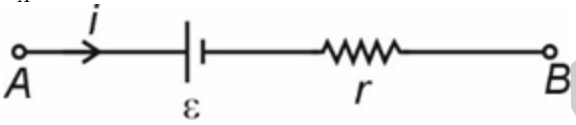
(3) **Answer :** (2)
Solution:
χ_m of a diamagnetic substance is independent of temperature.

(4) **Answer :** (4)
Solution:
$B_0 = 510 \times 10^{-9}$ T, E_0 = ?
$C = \frac{E_0}{B_0}$
$E_0 = CB_0 = 3 \times 10^8 \times 5.1 \times 10^{-7} = 153$ N/C

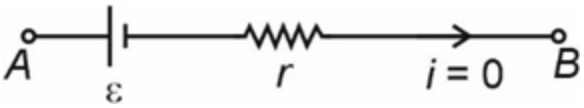
(5) **Answer :** (1)
Solution:



$V_A - V_B = \varepsilon - \text{ir}$ As a source.



1. $V_A - V_B = \varepsilon + \text{ir}$ As a load.



If, $i = 0$, $V_A - V_B = \varepsilon$.
Statement-3 is wrong.

(6) **Answer :** (2)
Solution:
Battery is disconnected.
Therefore charge cannot flow.
i.e., Charge → Q → Same
⇒ Dielectric plate is inserted, so
$C_{new} = KC$; $K > 1$
∴ C → Increases
⇒ Electric field $E_{new} = \frac{\sigma}{\varepsilon_0 K}$
⇒ E → Decreases
⇒ Energy density = $\frac{\text{Energy}}{\text{Volume}} = \frac{Q^2}{2CV}$
Since C↑, therefore energy density↓

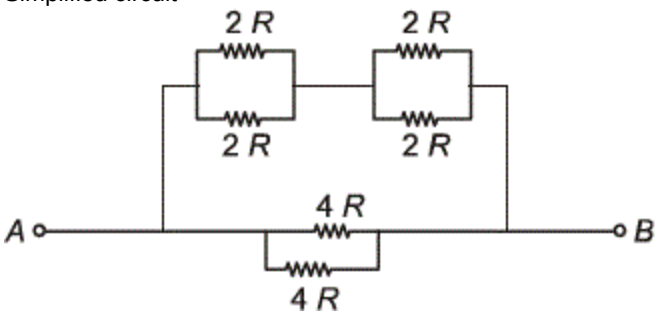
(7) **Answer :** (1)
Solution:

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$X_{COM} = \frac{2m+4m}{3m} = 2$

$Y_{COM} = \frac{4m+2m}{3m} = 2$

(8) **Answer :** (4)
Hint:
Circuit analysis
Solution:
Simplified circuit



$R_{net} = \frac{(R+R)(2R)}{2R+(R+R)} = \frac{4R^2}{4R} = R$

(9) **Answer :** (1)
Solution:
$X_C = X_L$ $\rightarrow$resonance condition
so, $V_R = V_{Source}$ = 300 V.

(10) **Answer :** (1)
Solution:
Velocity → $[LT^{-1}]$
Acceleration → $[LT^{-2}]$
Force → $[MLT^{-2}]$
Pressure → $[ML^{-1}T^{-2}]$

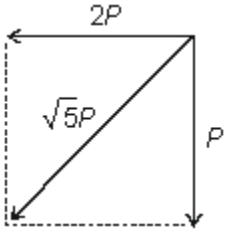
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(11) **Answer :** (4)
Solution:
Answer (4)
x = A sin ωt
A = 1 cm
$\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = \frac{\pi}{4}$ rad/s
x = sin $\frac{\pi t}{4}$
a = $-\frac{\pi^2}{16}\sin\frac{\pi t}{4}$
At $t = \frac{4}{3}$ s
$a = \frac{-\pi^2}{16}\sin\frac{\pi}{4}\left(\frac{4}{3}\right)$
$a = -\frac{\pi^2}{16}\left(\frac{\sqrt{3}}{2}\right)$
$a = -\frac{\pi^2\sqrt{3}}{32}$

(12) **Answer :** (3)
Solution:
Area of graph × 2 = work done
$W = 2\times\left[\frac{1}{2}\times 8\times 4 - \frac{1}{2}\times 2\times 1\right]$ = 2[16 – 1] = 30 J

(13) **Answer :** (2)
Solution:
$\frac{15}{12} = \frac{R}{4}$ $\quad\therefore\quad R = 5\Omega$
$5 = \frac{10x}{10+x}$ $\quad\therefore\quad 50+5x = 10x$
$50 = 5x$ or $x = 10\ \Omega$

(14) **Answer :** (3)
Solution:



$P_{\text{net}} = \sqrt{(2P)^2 + P^2} = \sqrt{5}P = \sqrt{5}qa$

(15) **Answer :** (1)
Solution:
Distance between two first minima is width of central maxima.
$W = \frac{2\lambda D}{d}$
Where *d* is width of slit.

(16) **Answer :** (4)
Solution:
$v' = 0.6\sqrt{2gh}$
$h' = \frac{v^2}{2g} = \frac{(0.6)(0.6)2gh}{2g}$
$h' = 0.36 \times 1$ m
= 0.36 m

(17) **Answer :** (1)
Solution:
Stress = $Y\ \alpha\ \Delta T$
Tension = $Y\ \alpha\ \Delta TA$
= ($2 \times 10^{11} \times 10^{-5} \times 200 \times 1 \times 10^{-4}$)
= ($2 \times 10^2 \times 200$)
= 4×10^4 N

(18) **Answer :** (4)
Solution:
$P_1 = -3.0$ D, $P_2 = 1.0$ D
$P = P_1 + P_2 = -3.0 + 1.0 = -2$ D
$f = \frac{1}{P} = \frac{1}{-2} = -0.5$ m or -50 cm

(19) **Answer :** (4)
Solution:
In uniform circular motion, the speed of particle is constant.
Hence, Kinetic energy also remains unchanged.

(20) **Answer :** (3)
Solution:
$U_i = \frac{-GMm}{R}$, $U_f = \frac{-GMm}{3R}$
$\Delta U = \frac{-GMm}{3R} + \frac{GMm}{R} = \frac{GMm}{R} \times \frac{2}{3}$
$\frac{2}{3}\frac{m}{R} + gR^2 = \frac{2mgR}{3}$

(21) **Answer :** (4)
Solution:
Significant figure in all measured value is 4.

(22) **Answer :** (2)
Solution:
$U = \frac{kq_1q_2}{r_1} + \frac{kq_1q_2}{r_2} + \ldots$
$U = kq_1q_2\left(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \ldots\infty\right)$

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$= 9 \times 10^9 \times 1 \times 2 \times 10^{-12} \left(\frac{1}{1-\frac{1}{2}}\right)$

$= 36 \times 10^{-3}$

$= 3.6 \times 10^{-2}$ J

(23) **Answer :** (3)
Solution:
$B = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$

(24) **Answer :** (2)
Solution:
$I_{req} + I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

(25) **Answer :** (1)
Solution:
$\delta = i_1 + i_2 - A$
$\delta = 40° + 40° - 60° = 20° = \frac{\pi}{9}$

(26) **Answer :** (3)
Hint:
At maximum height, $\frac{dy}{dt} = 0$
Solution:
$y = 4t - 2t^2$
At maximum height, $\frac{dy}{dt} = 0$
$\frac{dy}{dt} = 4 - 4t = 0 \quad \Rightarrow t = 1$ s
So, $x = 3$
$t = 3$ m

(27) **Answer :** (1)
Solution:
$^A_Z X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta \rightarrow {}^{A-4}_{Z-1}B* + {}^{4}_{2}He \rightarrow {}^{A-4}_{Z-1}B + \gamma$

(28) **Answer :** (3)
Solution:
Answer (3)
$T = 2\pi\sqrt{\frac{m}{k_{eq}}}$ where $\frac{1}{k_{eq}} = \frac{1}{k} + \underbrace{\frac{1}{2k}}_{\text{eq of spring in parallel}} = \frac{3}{2k}$

$\Rightarrow k_{eq} = \frac{2k}{3}$

(29) **Answer :** (4)
Solution:
Hint : Limiting friction between *B* and *C* = *mg*
Sol. : Acceleration of system *a* = $\frac{F}{6m}$
Normal reaction between *B* and *C*, *N* = 2*ma*
$N = \frac{F}{3}$
Now *B* will not slide downward if
μ*N* ≥ *mg*
$\mu\frac{F}{3} \geq mg$
$F \geq \frac{3mg}{\mu}$
$F_{min} = 6mg$

(30) **Answer :** (4)
Solution:
$\vec{a} \perp \vec{B}$
$\vec{a} \cdot \vec{B} = 0$

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$-6 + 2x = 0$
$2x = 6$
$x = 3$

(31) Answer : (3)
Solution:
As the velocity will be minimum at point A, hence magnitude of linear momentum minimum at point A.

(32) Answer : (1)
Solution:

$E = (-13.6 \text{ eV}) \frac{z^2}{n^2}$

$z = 2$; $n = 2$
$\therefore E = -13.6$ eV

(33) Answer : (4)
Solution:
$2Tl = mg$

$T = \frac{mg}{2l} = \frac{1.5 \times 10^{-2}}{2 \times 30 \times 10^{-2}} = 0.025$ N/m

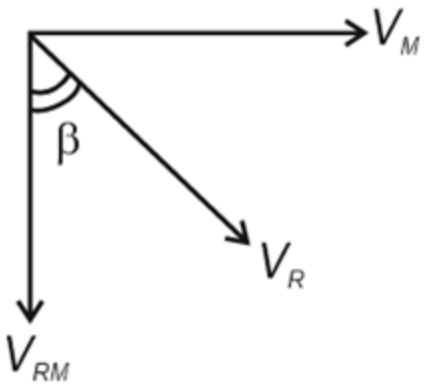
(34) Answer : (4)
Solution:
All electron emitted do not have same K.E; $(K.E)_{max} = hf - \phi_0$

(35) Answer : (2)
Solution:

$V_{avg} = \frac{3x}{\frac{x}{10} + \frac{x}{20} + \frac{x}{30}}$

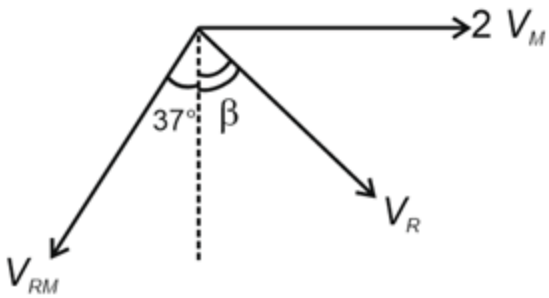
= 16.4 m/s

(36) Answer : (1)
Solution:



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$\Rightarrow \quad V_R \sin\beta = 6$ km/h



$\Rightarrow \quad \tan 37^\circ = \frac{6}{V_R \cos\beta}$

$V_R \cos\beta = \frac{4 \times 6}{3} = 8$ km/h

$V_R = 10$ km/h

(37) Answer : (2)
Hint:

Use $M = \frac{\phi}{i}$ and $\phi = \int \vec{B} . \vec{dA}$

Solution:

Since $\phi_B = \int B \cdot c dr = \int_a^{a+b} \frac{\mu_0 i}{2\pi r} c dr$

$= \frac{\mu_0 ic}{2\pi} |\ln r|_a^{a+b}$

$= \frac{\mu_0 ic}{2\pi} \ln\left(\frac{a+b}{a}\right)$

$M = \frac{\phi}{i} = \frac{\mu_0 c}{2\pi} \ln\left(1 + \frac{b}{a}\right)$

(38) Answer : (1)
Solution:
Inside the cavity of a uniformly charged sphere, electric field is uniform. Therefore, charge experiences the same force at all positions inside the cavity.

(39) Answer : (1)
Solution:
The boolean expression of this circuit is $Y = A \cdot B$ i.e. this circuit resembles AND gate.

(40) Answer : (1)
Solution:

$\theta = \omega_0 t + \frac{1}{2}\alpha\, t^2$

$200 = 0 \times 5 + \frac{1}{2}\alpha \times 25$

$\alpha = 16$ rad/s^2

(41) Answer : (2)
Solution:

$\frac{\beta}{\lambda}(\mu - 1)t = \frac{3}{2}\beta$

$\lambda = \frac{2(\mu-1)t}{3}$

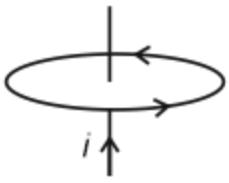
(42) Answer : (2)
Solution:
In equation $y(x,\ t) = \frac{1.8}{(2x+5t)^2+5}$

As coefficient of x and t are of same sign the wave is travelling in negative x direction.

Wave speed = $\frac{|\text{coefficient of } t|}{|\text{coefficient of } x|} = \frac{5}{2} = 2.5$ m/s

⇒ Wave will travel 5 m in 2 s with speed of 2.5 m/s
At maximum displacement y should be at its maximum i.e. $2x + 5t$ should be equal to zero
⇒ $y_{\max} = \frac{1.8}{5} = 0.36$ m

(43) Answer : (4)
Solution:



All the statements are correct.

(44) Answer : (2)
Hint:

Speed of longitudinal waves in a gas is independent of pressure at constant temperature. So $v \propto P^0$.
Solution:

Speed of longitudinal waves in a gas is independent of pressure at constant temperature. So $v \propto P^0$.

(45) Answer : (4)

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Solution:
Answer (4)

$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$

CHEMISTRY

(46) Answer : (4)
Hint:

Mole of molecule/atom = $\frac{\text{Mass of molecule / atom}}{\text{Molar mass of molecule / atom}}$

Solution:

Mole of He = $\frac{2x}{4}$

Mole of N_2 = $\frac{x}{28}$

$\frac{\text{Mole of He}}{\text{Mole of } N_2} = \frac{\frac{2x}{4}}{\frac{x}{28}} = \frac{2x}{4} \times \frac{28}{x} = 14:1$

(47) Answer : (1)
Hint:

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

Solution:

Mole of H_2 = $\frac{2.24}{22.4} = 0.1$

Mole of Zn required = 0.1

Mass of Zn required = 0.1 × 65 = 6.5 g

(48) Answer : (2)
Hint:

Electron is more tightly bound to the nucleus in smallest allowed orbit.

Solution:

electron has more negative energy in n = 1 than n = 5.

∴ electron is tightly bound to nucleus in n = 1.

For 'l'subshell, maximum number of electrons are $4l+2$

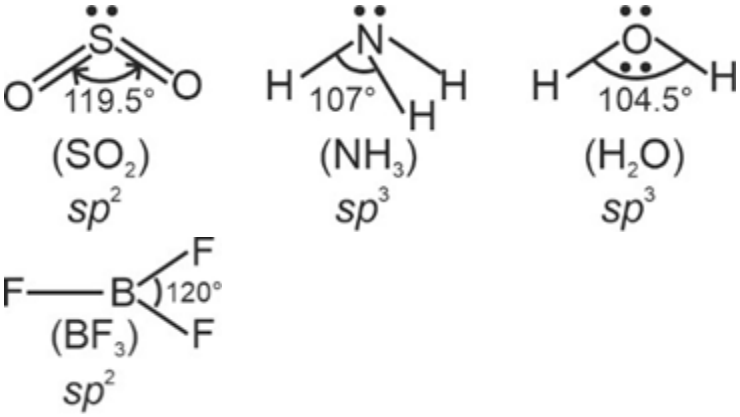
(49) Answer : (2)
Hint:

According to Pauli's exclusion principle; no two electrons in an atom can have the same set of four quantum numbers.

Solution:

It follows only Pauli's exclusion principle. It violates Hund's rule of maximum multiplicity and Aufbau principle.

(50) Answer : (4)
Solution:



(51) Answer : (2)
Solution:

The hybridisation of N is sp^3 in both NH_3 and NH_4^+.

(52) Answer : (1)

Hint:

W = – 2.303 nRT log $\frac{V_f}{V_i}$

Solution:

W = – 2.303 × 2 × 8.314 × 300 × log $\frac{50}{10}$

= – 2.303 × 2 × 8.314 × 300 × 0.7

= – 8.04 kJ

(53) Answer : (4)

Solution:

$C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6\,CO_2(g) + 6H_2O(l)$

$\Delta H = \Sigma(\Delta_f H)_P - \Sigma(\Delta_f H)_R$

$(\Delta_f H)_{CO_2} = (\Delta_c H)_{C(graphite)}$

$\Delta_r H = -2400 - 1800 + 1300 = -2900\ kJ$

$\Delta U = \Delta H - \Delta n_g(RT)$

$\Delta U = -2900 - 0(RT) = -2900\ kJ\ mol^{-1}$

$= \frac{-2900}{180}\ kJ\,g^{-1} = -16.11\ kJ\,g^{-1}$

(54) Answer : (1)

Solution:

At equivalence point number of equivalents of base = number of equivalents of acid

$2.5 \times \frac{2}{5} \times 1 = V \times \frac{2}{15} \times 1$

$V = \frac{2.5}{5} \times 15 = 2.5 \times 3 = 7.5\ mL$

BOH + HCl → BCl + H_2O

$\Rightarrow\ 2.5 \times \frac{2}{5} \rightarrow \frac{2}{15} \times 7.5$

1 mmol 1 mmol

\- - 1 mmol

Concentration of $BCl = \frac{1\ mmol}{10\ mL} = 10^{-1}$

$pH = 7 - \frac{1}{2}(pK_b + logc)$

$pH = 7 - \frac{1}{2}(12 + \log 10^{-1})$

$= 7 - \frac{1}{2}(12 - 1)$

$= 7 - \frac{1}{2}(11)$

pH = 7 – 5.5

pH = 1.5

(55) Answer : (3)

Hint:

Exothermic reactions are favoured at low temperature.

Solution:

Addition of inert gas at constant volume has no effect on equilibrium state.

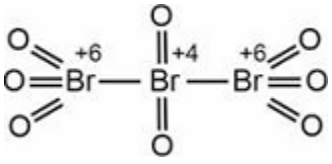
When we decrease the pressure, reaction will shifts towards that side which has higher number of gaseous moles.

(56) Answer : (2)

Hint:

For covalent bond, high electronegative element gets negative oxidation state and less electronegative element gets positive oxidation state.

Solution:



Average oxidation state of Br = $\frac{+16}{3}$

(57) Answer : (4)

Hint:

CO forms a stable complex with haemoglobin, this found to be poisonous.

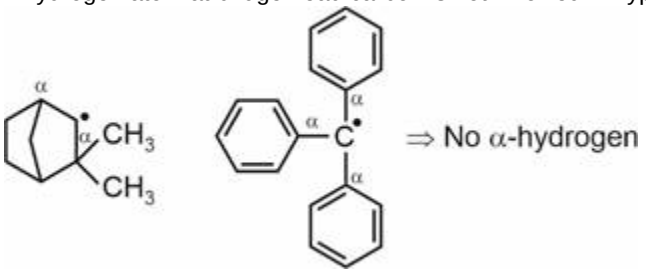
Solution:

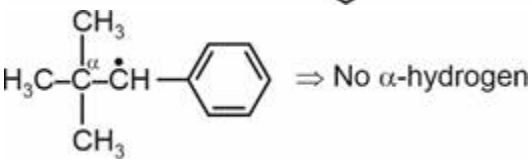
Zeolites are widely used as a catalyst in petrochemical industries for cracking of hydrocarbons and isomerisation.

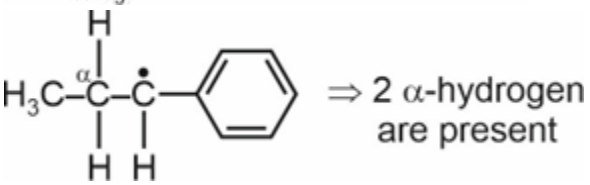
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The highly poisonous nature of CO arises due to its ability to form a complex with haemoglobin which is about 300 times more stable than the oxygen-haemoglobin complex.
Being heavy and non-supporter of combustion CO_2 is used as fire extinguisher.

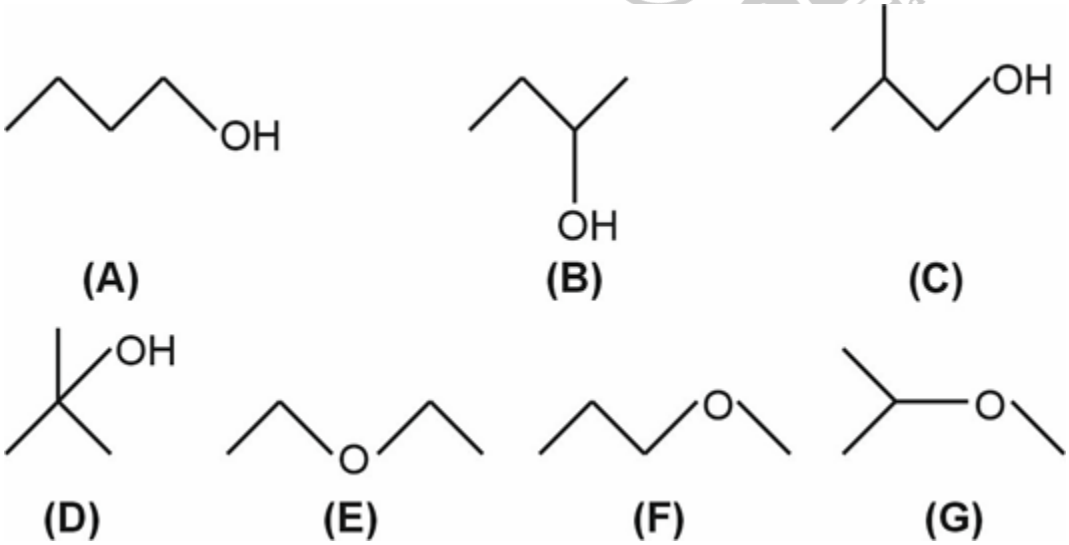
(58) Answer : (2)
Solution:
• If hydrogen atom is present at α carbon to the free radical then hyper conjugation is possible.
• Hydrogen atom at bridge head carbon is not involved in hyperconjugation.







(59) Answer : (1)
Solution:
Total 8 isomers are possible for molecule formula $C_4H_{10}O$



• (B) exists as a pair of enantiomers.

(60) Answer : (4)
Solution:
233 g of $BaSO_4$ contains 32g of S
0.466 g of $BaSO_4$ contains $\frac{32\times0.466}{233}$ g of S
% of sulphur = $\frac{32\times0.466\times100}{233\times0.25} = 25.6\%$

(61) Answer : (2)
Solution:
Cyclic, planar species having (4n + 2)π electrons in conjugation are aromatic in nature.

(62) Answer : (2)

Solution:

$$2\,CH_4 + O_2 \xrightarrow[100\ atm]{Cu,\ 523\,K} 2\,CH_3\,OH$$

$$CH_4 + H_2O \xrightarrow[\Delta]{Ni} CO + 3H_2$$

$$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2\,O_3} HCHO + H_2O$$

$$2\,CH_3\,CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3\,COO)_2\,Mn} 2\,CH_3\,COOH + 2H_2O$$

(63) Answer : (4)
Solution:

$\pi_{Na_2\,SO_4} = \pi_{glucose}$

$i_1c_1 = i_2c_2, \quad \begin{pmatrix} i_1 = 1 + 2\alpha \\ i_2 = 1 \end{pmatrix}$

0.004 (1 + 2α) = 0.0108
1 + 2α = 2.7
α = 0.85
% Dissociation = α × 100
= 85%

(64) Answer : (3)
Hint:

$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

Solution:
To get 1 mol of O_2 from water, we need to pass 4F charge

(65) Answer : (1)
Solution:

Cell reaction, Ni(s) + $2H^+$(aq) → Ni^{2+}(aq) + H_2(g)

$E^o_{cell} = E^o_{H^+|H_2} - E^o_{Ni^{2+}|Ni} = 0.236$

$$E_{cell} = E^o_{cell} - \frac{0.0591}{n}\log_{10}\frac{\left[p_{H_2}\right]\left[Ni^{2+}\right]}{\left[H^+\right]^2}$$

$$\Rightarrow\ 0 = 0.236 - \frac{0.0591}{2}\log_{10}\frac{1\times1}{\left[H^+\right]^2}$$

$$\Rightarrow\ -0.236 = -\frac{0.0591}{2}\times 2\times pH \quad \left(\because \log_{10}\frac{1}{[H^+]} = pH\right)$$

$$\Rightarrow\ pH = \frac{0.236}{0.0591} = 4$$

(66) Answer : (2)
Hint:
Rate of reaction = $k[A]^x[B]^y[C]^z$
Solution:

$5\times10^{-5} = k(0.01)^x(0.005)^y(0.01)^z$ ....(1)
$5\times10^{-5} = k(0.02)^x(0.005)^y(0.01)^z$ ....(2)
$2.5\times10^{-5} = k(0.01)^x(0.01)^y(0.01)^z$ ....(3)
$20\times10^{-5} = k(0.01)^x(0.005)^y(0.02)^z$ ....(4)
By solving (1) and (2), x = 0
By solving (1) and (3), y = –1
By solving (1) and (4), z = 2
∴ Rate of reaction $= k[A]^0[B]^{-1}[C]^2$

(67) Answer : (1)
Hint:

$k = Ae^{-E_a/RT}$

$$\log k = \log A - \frac{E_a}{2.303\,RT}$$

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Solution:

$\log k = \log A - \frac{E_a}{2.303\,RT}$ ...(i)

$\log k = 5 - \frac{3000}{T}$ ...(ii)

On comparing equation (i) and (ii)

logA = 5

$A = 10^5$

$\frac{E_a}{2.303R} = 3000$

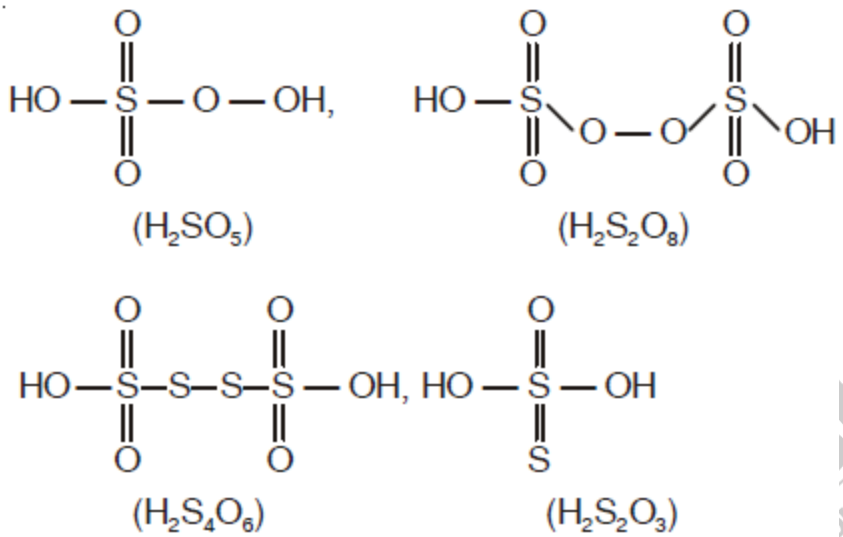
E_a = 57.44 kJ/mol

(68) Answer : (1)

Hint:

Peroxy compound contains O—O linkage.

Solution:



(69) Answer : (2)

Solution:

Hydrides	NH_3	PH_3	AsH_3	SbH_3
Bond angle	107.8°	93.6°	91.8°	91.3°

(70) Answer : (1)

Hint:

As the atomic number increases in actinoids, the ionic radii decreases due to increase in effective nuclear charge.

Solution:

The correct order is

$Am^{3+} < Pu^{3+} < Np^{3+} < U^{3+} < Ac^{3+}$

(71) Answer : (1)

Solution:

$[Co(NH_3)_4Cl_2]Cl + AgNO_3 \rightarrow AgCl$ (1 mole) + $[Co(NH_3)_4Cl_2]^+$

$[Co(NH_3)_3Cl_3] + AgNO_3 \rightarrow$ No precipitate of AgCl

$[Co(NH_3)_5Cl]Cl_2 + 2AgNO_3 \rightarrow 2AgCl + [Co(NH_3)_5Cl]^{+2}$

(72) Answer : (4)

Hint:

cis-$[CrCl_2(ox)_2]^{3-}$ is optically active molecule.

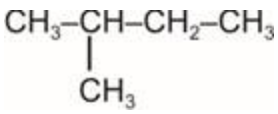
Solution:

cis-$[CrCl_2(ox)_2]^{3-}$ lacks plane of symmetry and centre of symmetry, so, it is a chiral molecule.

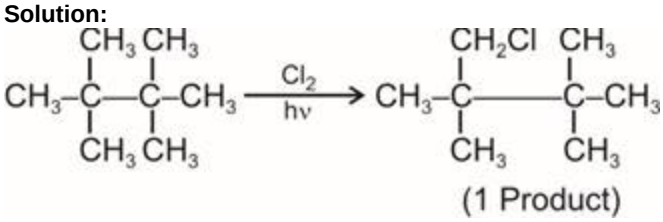
(73) Answer : (3)

Hint:

Isopentane is

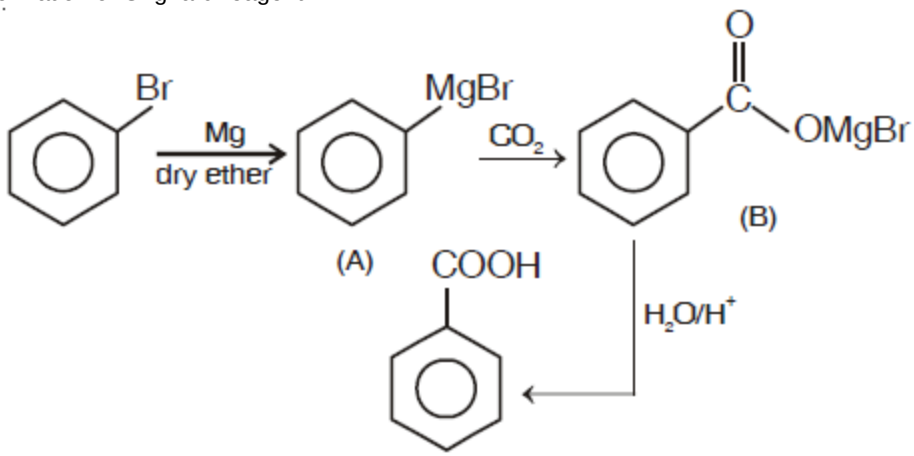


Solution:



2,2,3,3-tetramethylbutane
∵ All 18 hydrogen atoms are equivalent
∴ Only one monochlorinated product is obtained.

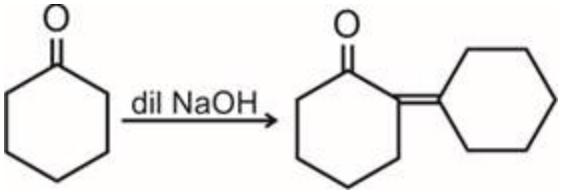
(74) **Answer :** (4)
Solution:
Formation of Grignard reagent.



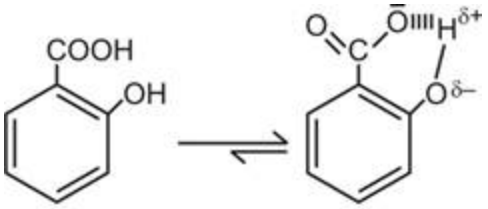
(75) **Answer :** (2)
Hint:
High boiling points of alcohols because of involving hydrogen bonding.
Solution:

Molecules	Boiling point (K)
Ethanol	351 K
Methoxy methane	248 K
Propane	231 K
Butan-1-ol	390 K

(76) **Answer :** (2)
Solution:



(77) **Answer :** (4)
Hint:
More stable the conjugate base, stronger is the acid.
Solution:



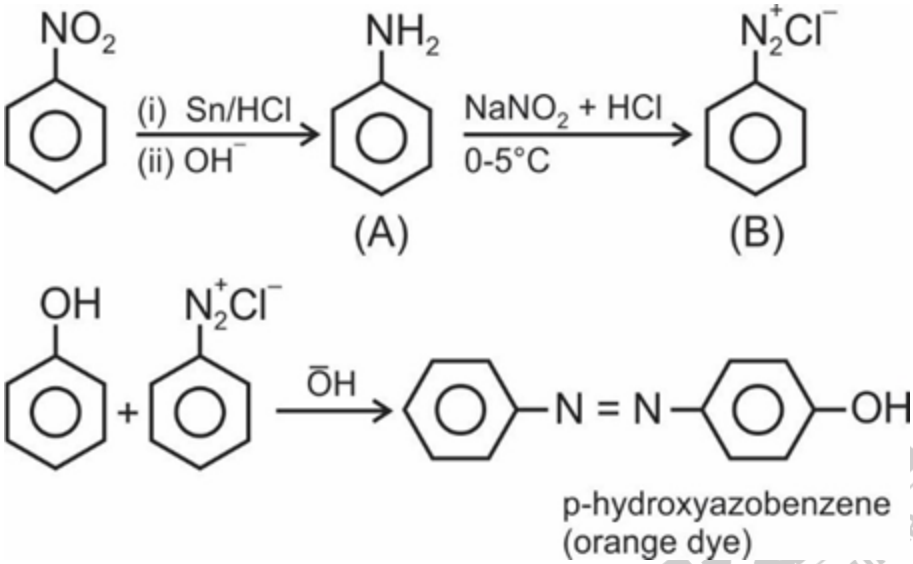
Conjugate base is stabilised by hydrogen bond hence O – hydroxybenzoic acid is strongest acid

(78) **Answer :** (1)

Hint:

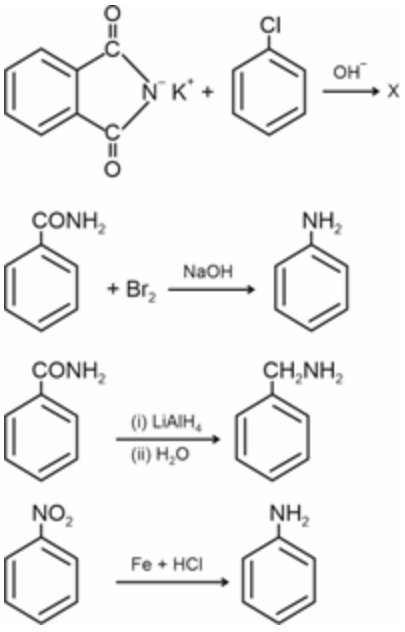
Nitro benzene on reduction with Sn/HCl gives aniline.

Solution:



(79) **Answer :** (3)

Solution:

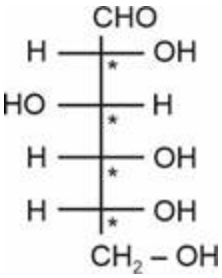


(80) **Answer :** (1)

Hint:

Any carbon atom attached to four different groups is a chiral (stereo) centre.

Solution:



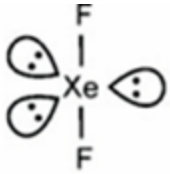
* → Chiral centre or stereo centre
∴ No. of stereo centres = 4

(81) **Answer :** (2)
Solution:
Nessler's reagent is K_2HgI_4,
$2K_2HgI_4 + NH_3 + 3KOH \rightarrow HgO{\cdot}Hg(NH_2)I + 7KI + 2H_2O$

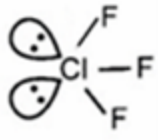
(82) **Answer :** (2)
Solution:

Molecular Orbitals	Number of Nodes
$\pi 2p_x$	1
$\pi^* 2p_x$	2
$\sigma^* 1s$	1

(83) **Answer :** (2)
Solution:



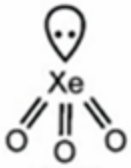
(XeF_2)
(Linear)



(ClF_3)
(Bent-T-shape)



(SF_4)
(See-saw)



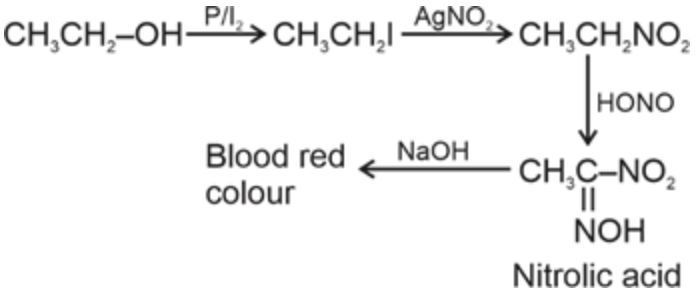
(XeO_3)
(Pyramidal)

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(84) **Answer :** (2)
Solution:
Hint: Lactose is composed of glucose and galactose.
Sol.: $Lactose \xrightarrow{hydrolysis} \beta - D - Galactose + \beta - D - Glu\cos e$

(85) **Answer :** (2)
Hint:
For concentrated H_2SO_4, oxidation of sulphate ions take place at anode.
Solution:
At anode: $2SO_4^{2-}(aq) \rightarrow S_2O_8^{2-}(aq) + 2e^-$

(86) **Answer :** (1)
Solution:
Hint : Primary and secondary alcohols give colour in Victor Meyer Test.
Sol. :



(87) **Answer :** (1)
Solution:

- The correct electronic configuration of berkelium(Bk) is $[Rn]5f^{9}7s^{2}$
- V_2O_5 catalyses the oxidation of SO_2 in the manufacture of sulphuric acid
- Nickel complexes are useful in the polymerisation of alkynes
- Mischmetall consists of a lanthanoids metal (~95%), Fe(~5%) and traces of S, C, Ca and Al.

(88) **Answer :** (4)
Solution:

Group-I – dilute HCl

Group-II – H_2S + dilute HCl

Group-III – NH_4OH + NH_4Cl

Group-IV – H_2S + NH_4OH

(89) **Answer :** (3)
Hint:
Metals have low electronegativity while non-metals have high electronegativity.
Solution:
Electronegativity is directly related to that non-metallic properties of elements. Thus, the increase in electronegativities across a period is accompanied by an increase in non metallic properties.
Electronegativity is a non-measurable quantity. Pauling assigned arbitrarily value of 4.0 to fluorine.

(90) **Answer :** (2)
Hint:
Chlorine has maximum negative electron gain enthalpy.
Solution:
N is more electronegative than P.

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BIOLOGY

(91) **Answer :** (3)
Solution:
The correct sequence of taxonomic hierarchy for the housefly, starting from the lowest rank, is:
(iii) *Musca* → (v) Muscidae → (ii) Diptera → (vi) Insecta → (i) Arthropoda → (iv) Animalia

(92) **Answer :** (4)
Solution:
Prions cause diseases like Bovine spongiform encephalopathy commonly called mad cow disease in cattle and Cr-Jacob disease in humans (CJD).
Potato spindle tuber disease is cased by viroids.

(93) **Answer :** (3)
Solution:
Euglena is chlorophyllous, flagellate protist.

(94) **Answer :** (4)
Solution:
Mycoplasma is the smallest wall-less moneran, which is mostly parasitic.

(95) **Answer :** (3)
Solution:

Antheridium and archegonium are male and female sex organs respectively.

(96) Answer : (4)
Solution:
The given plant is *Laminaria.* It belongs to Phaeophyceae in which major pigments are chl *a*, *c* and fucoxanthin.

(97) Answer : (1)
Solution:
Marchantia is a bryophyte and its plant body is differentiated into thallus or foliose structure and rhizoids.

(98) Answer : (3)
Hint:
Maturation zone is proximal to region of elongation.
Solution:
In meristematic cells, the cells are thin walled. Root cap is a multicellular thimble like structure.
Elongation zone has mature cells which do not divide.

(99) Answer : (2)
Solution:
(b) Many ovules
(c) Berry or capsule

(100) Answer : (2)
Hint:
Valvate aestivation of calyx is present in China rose.
Solution:
Twisted aestivation of corolla and monoadelphous androecium is found in China rose.

(101) Answer : (3)
Solution:
Xylem and phloem are situated at the same radius of vascular bundles in conjoint type

(102) Answer : (3)
Solution:
All the given features are true for dicot leaf.

(103) Answer : (3)
Hint:
Lipid molecules of plasma membrane have polar head and non-polar tail.
Solution:
Polar head is hydrophilic in nature and it interacts with water. Rest all the statements regarding plasma membranes are true.

(104) Answer : (1)
Solution:
(a) Capsule of bacteria is made up of polysaccharides and D-glutamic acid. It provides virulent property to the cell and helps to hide from host's immune system.
(b) Ribosomes are present in all living cell, both prokaryotes & eukaryotes
(c) Vacuoles are not polymorphic, they are found in plant cells.
(d) Peroxisomes are found in both plant and animal cells. They are involved in detoxification.

(105) Answer : (3)
Solution:
Amyloplast stores starch and it is a colourless plastid.

(106) Answer : (1)
Solution:
Smooth muscle fibres are uninucleated and fusiform in shape.

(107) Answer : (1)
Solution:
Microvilli is present for increase in surface area.

(108) Answer : (1)
Solution:
Brood pouch in male cockroach is bounded dorsally by 9th and 10th terga and ventrally by 9th sternum.

(109) Answer : (2)
Solution:
In frogs, the mouth opens into the buccal cavity that leads to oesophagus through pharynx. Oesophagus is a short tube that opens into the stomach.

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(110) Answer : (2)
Solution:
Over two-thirds of all named species on earth are arthropods. The body of arthropods is covered by chitinous exoskeleton.

(111) Answer : (3)
Solution:
Chondrichthyes lack operculum and air bladder.

(112) Answer : (1)
Solution:
Cloaca is present in most fishes, non-mammalian tetrapods and monotreme mammals. Digestive tract and urogenital passages open into cloaca.

(113) Answer : (2)
Solution:
Palmitic acid has 16 carbon atoms including the carboxyl carbon.

(114) Answer : (2)
Solution:
The essential chemical components of many coenzymes are vitamins *e.g*. coenzyme nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin.

(115) Answer : (3)
Solution:
The difference in average energy content of 'S' from that of transition state is called 'activation energy'. Enzymes eventually bring down this energy barrier making the transition of 'S' to 'P' more easy.

(116) Answer : (3)
Solution:
Nucleolus is not surrounded by any membrane.

(117) Answer : (2)
Solution:
Centrifugal cytokinesis is seen in plants and its assisted by cell plate.

(118) Answer : (1)
Solution:
(i) Anaphase-I
(ii) Pachytene
(iii) Metaphase-I
(iv) Anaphase-II
Correct sequence will be (ii) → (iii) → (i) → (iv)

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(119) Answer : (1)
Solution:
Chromosomes are most condensed and distinctly visible during metaphase. Cytokinesis phase marks the end of cell-division.

(120) Answer : (2)
Hint:
In C_4 plants, Calvin cycle does not occur in mesophyll cells.
Solution:
In C_4 plants, Calvin cycle occurs in bundle sheath cell where RuBisCO is present.
ATP for Calvin cycle comes from light reactions of photosynthesis that occur in chloroplasts.

(121) Answer : (2)
Solution:
Carboxylation is the most crucial step of the Calvin cycle where CO_2 is utilised for the carboxylation of RuBP. This reaction is catalysed by the enzyme RuBP carboxylase which results in the formation of 2 molecules of 3-PGA.
The regeneration steps require one ATP for phosphorylation to form RuBP.
To produce one molecule of glucose, 6 turns of Calvin cycle are required.

(122) Answer : (4)
Solution:
Pyruvic acid is the common molecule/link between the respiratory-mediated breakdown of carbohydrates and proteins.

(123) Answer : (4)
Solution:
Oxidative phosphorylation occurs in mitochondria during aerobic respiration.

(124) Answer : (2)

Solution:
• Auxin controls the xylem differentiation
• Abscisic acid is responsible for closure of stomata.
• Gibberellins are used to increase the size of apple
• Ethylene is a ripening agent.

(125) Answer : (3)
Solution:
Ethylene induces rapid elongation of internode and petiole in rice seedlings in water.

(126) Answer : (4)
Solution:
Embryo sac — Female gametophyte

(127) Answer : (1)
Solution:
Geitonogamy is the transfer of pollen grains from anther to the stigma of another flower of the same plant.

(128) Answer : (2)
Solution:
Single cotyledon of monocot embryo is called scutellum.
Endosperm = 3n, perisperm = 2n
Hollow foliar structure enclosing shoot apex and leaf primordia is coleoptile.
Epiblast - remains of second cotyledon in monocots.

(129) Answer : (3)
Solution:
Down's syndrome occurs due to trisomy while turner's due to monosomy. Flower colour in snapdragon shows incomplete dominance. Test cross show 1 : 1 ratio in case of completely linked genes in dihybrid cross.

(130) Answer : (2)
Solution:
The given pedigree represents autosomal recessive trait *e.g.*, Sickle cell anaemia.

(131) Answer : (4)
Hint:
Asthma is caused by allergens.
Solution:
Asthma usually occurs due to an allergic reaction to foreign substances that affect the respiratory tract. The major cause of emphysema is excessive cigarette smoking that causes damage to alveolar walls. Occupational respiratory disorders occur due to continuous exposure to harmful substances such as, fine fibres of jute, cotton, silica dust, *etc*.

(132) Answer : (4)
Hint:
Think about basement membrane
Solution:
At the tissue site, where pCO_2 is high due to catabolism, CO_2 diffuses into RBCs and forms HCO_3^- and H^+ in the presence of enzyme carbonic anhydrase. Diffusion membrane is formed by two cellular layers and an acellular basement substance in between them.

(133) Answer : (3)
Hint:
Identify CAD
Solution:

Angina	Acute chest pain
Cardiac arrest	Heart stops beating
Atherosclerosis	Lumen of arteries become narrower due to deposits of calcium, fat, cholesterol, *etc*.
Heart attack	Heart muscle damaged by an inadequate blood supply.

(134) Answer : (3)
Solution:
These animals live on both land and water.
Crocodiles are reptiles with completely divided four chambered heart. Amphibians have three chambered heart.

(135) Answer : (2)
Solution:
SA node is located in the upper right corner of the right atrium.

(136) Answer : (4)

Solution:

Nitrogenous wastes are absent in dialysing fluid which is why these substances freely move out clearing the blood.

(137) Answer : (3)

Hint:

Insulin deficiency.

Solution:

Presence of glucose and ketone bodies in urine are indicative of diabetes mellitus.

(138) Answer : (3)

Solution:

200 mL CO_2 per minute

$\therefore$ 200 × 60 = 12000 mL/hour

$\therefore$ 12 L in 1 hour.

(139) Answer : (3)

Solution:

Myasthenia gravis: Auto-immune disorder affecting neuro-muscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

Osteoporosis: Age-related disorder characterised by decreased bone mass and increased chances of fractures. Decreased levels of estrogen is a common cause.

Gout: Inflammation of joints due to accumulation of uric acid crystals.

(140) Answer : (1)

Solution:

In humans, each meromyosin has two important parts, a globular head with a short arm and a tail; the former being called the heavy meromyosin (HMM) and the latter, the light meromyosin (LMM).

The HMM component, *i.e.*, the head and the short arm projects outwards at regular distance and angle from each other from the surface of a polymerised myosin filament, and is known as cross arm.

The globular head is an active ATPase enzyme and has binding sites for ATP and active sites for actin.

(141) Answer : (4)

Solution:

Na^+-K^+ pump transports 3 Na^+ outside for 2K^+ into the cell.

(142) Answer : (2)

Solution:

Based on the number of axon(s) and dendrite(s), the neurons are divided into three types, *i.e.*,

(A) Multipolar (with one axon and two or more dendrites; found in the cerebral cortex)

(B) Bipolar (with one axon and one dendrite, found in the retina of eye)

(C) Unipolar (cell body with one axon only; found usually in the embryonic stage)

(143) Answer : (2)

Solution:

Addison's disease	–	Underproduction of hormones from adrenal cortex
Graves' disease	–	Hyperthyroidism
Diabetes mellitus	–	Prolonged hyperglycemia
Acromegaly	–	Hyperproduction of growth hormone in adults

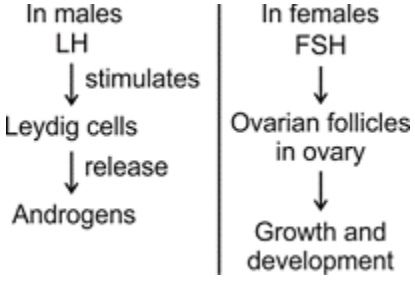
(144) Answer : (1)

Solution:

Cortisol is a glucocorticoid which is involved in carbohydrate metabolism.

(145) Answer : (1)

Solution:



(146) Answer : (1)

Hint:

Labia majora extend down from this tissue.

Solution:

The uppermost cushion of fatty tissue covered with skin and pubic hairs is called mons pubis. Labia majora extend down from the mons pubis and surrounds the vaginal opening. Clitoris is erectile tissue present at the upper junction of two labia minora above the urethral opening. Hymen partially covers the opening of the vagina.

(147) Answer : (2)

Solution:

Fully developed foetus and placenta initiate signals for parturition.

(148) Answer : (1)

Solution:

Lactiferous ducts open directly on nipples.

(149) Answer : (2)

Solution:

Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cells (oogonia) are formed within each fetal ovary; no more oogonia are formed and added after birth. These cells start division and enter into prophase-I of the meiotic division and get temporarily arrested at that stage, called primary oocytes.

(150) Answer : (3)

Solution:

Hymen can also be broken due to sudden fall or jolt or active participation in sports like horseback riding, *etc*. So, presence or absence of hymen is not a reliable sign of virginity.

• Oogenesis begins in embryonic development stages in human females.

(151) Answer : (4)

Solution:

Oral contraceptive pills inhibit the secretion of FSH and LH that prevent ovulation.

(152) Answer : (2)

Solution:

IUDs are intra uterine devices fitted into the uterus.

(153) Answer : (2)

Solution:

Silurian period is a part of Paleozoic era.

(154) Answer : (4)

Solution:

Homo neanderthalensis cover their body with animal hides and buried their dead.

(155) Answer : (3)

Solution:

Australian marsupials and placental mammals show convergent evolution.

(156) Answer : (3)

Solution:

Edward Wilson described the combined diversity at all levels of biological organisation. Robert May's estimate placed global species diversity at about 7 million. Alexander von Humboldt observed that within the region species richness increases with increasing explored area but up to a limit.

(157) Answer : (3)

Solution:

The recessive trait that causes sickle cell anaemia can be transmitted from parent to offspring when both the parents are the carrier for the gene.

(158) Answer : (2)

Solution:

Chromosomal theory of inheritance was proposed independently by Sutton and Boveri and experimentally verified by Morgan.

(159) Answer : (4)

Solution:

Regarding Watson and Crick, the pitch of the DNA helix is 3.4 nm.

(160) Answer : (4)

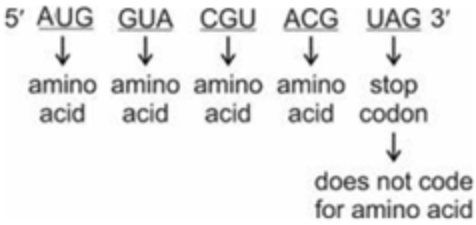
Solution:

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For the synthesis of DNA, replication fork is formed. The discontinuously synthesised DNA fragments on the template strand with polarity 5′ → 3′ join together by DNA ligase.

(161) Answer : (2)
Solution:
Degeneracy of codon refers that some amino acids are coded by more than one codons.

(162) Answer : (2)
Solution:
According to given question, stretch of DNA is
3′ TACC ATGCATGC ATC 5′ Template strand
5′ ATGG TACGTACG TAG 3′ Coding strand
Since, template strand is involved in transcription
Hence, the m-RNA produced will be



Hence, total amino acid produced will be four.

(163) Answer : (4)
Solution:
No chemicals are used in biocontrol of pests and disease.

(164) Answer : (3)
Hint:
Trichoderma polysporum produces cyclosporin A.
Solution:

Microbe	Product	Usage
Trichoderma polysporum	Cyclosporin A	Immunosuppressive agent
Monascus purpureus	Statins	Blood cholesterol- lowering agent
Streptococcus	Streptokinase	Clot buster
Acetobacter aceti	Acetic acid	Used in preparation of vinegar

(165) Answer : (4)
Solution:
(i) In interference competition, the feeding efficiency of one species is affected due to the inhibitory presence of the other.
(ii) Resources may not be limiting for competition to occur.

(166) Answer : (3)
Solution:
(i) *Penicillium* inhibiting the growth of surrounding bacteria is an example of Ammensalism (–, 0).
(ii) Abingdon tortoise and goat is an example of competition (–, –).

(167) Answer : (1)
Solution:

Plants (100,000 J)	→	Caterpillar (10,000 J)	→	Sparrow (1000 J)	→	Hawk (100 J)
T_1		T_2		T_3		T_4

(168) Answer : (1)
Solution:
NPP is the available biomass for consumption to heterotrophs.

(169) Answer : (4)
Solution:
Birds are around 1300 in number in Amazonian rainforest.

(170) Answer : (4)
Solution:

Communities with more biodiversity are more stable and more resistant to natural disturbances, man-made disturbances and invasion by alien species. Rich biodiversity decreases year-to-year variations in productivity of an ecosystem.

(171) Answer : (3)
Solution:
Streptococcus pneumoniae and *Haemophilus influenzae* are both pneumonia causing bacterial pathogens.

(172) Answer : (1)
Hint:
Identify an opiate narcotic drug.
Solution:
Given structure represents morphine. Hashish and marijuana are hallucinogenic drugs while cocaine is a stimulant.

(173) Answer : (3)
Solution:
Antibodies from the mother's milk are source of natural passive immunity.

(174) Answer : (2)
Solution:
Restriction enzymes cut DNA at specific locations and are therefore called molecular scissors.

(175) Answer : (4)
Solution:
Isolation, purification, formulation, packaging, storage and marketing of products are parts of downstream processing.

(176) Answer : (3)
Hint:
Microneedles are used.
Solution:
Biolistic/gene gun is suitable for plants. In method known as micro-injection, recombinant DNA is directly injected into nucleus of an animal cells.

(177) Answer : (1)
Solution:
Gene therapy was first performed in 1990. Humulin was synthesized in 1983.

(178) Answer : (4)
Solution:
Chitinase degrades fungal cell wall. Cellulase acts on plant cell wall. Pectinase acts on middle lamella present between plant cells.

(179) Answer : (3)
Hint:
This toxin generally acts at pH around 8.
Solution:
Bt toxin is proteinaceous in nature and it exists in inactive form but gets activated in midgut of various insects in alkaline medium.

(180) Answer : (2)
Solution:
Chain A and chain B of insulin were linked by disulphide bonds.

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